

HPC Project Report

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1 Introduction (??)

2 Assignment 1

2.1 Introduction and objectives

The objective of this project is to implement a simulator that models the propagation of a damped wave. The general partial differential equation describing the phenomenon in space as time varies is the following:

$$\frac{\partial^2 u}{\partial t^2} + \gamma \frac{\partial u}{\partial t} = c^2 \nabla^2 u \quad (1)$$

Where $u(x, y, t)$ represents the height of the wave on a two-dimensional plane over time, γ is the wave damping coefficient, and c is the wave propagation speed. The simulator calculates the wave height for each point in the coordinate space (x, y) as time t changes, with the ultimate goal of generating an MP4 video of the simulation.

2.2 Implementation

To perform the simulation, it is necessary to convert the continuous model into a discrete model suitable for implementation on a computer. For this purpose, a grid is introduced: space is divided into equally spaced points along the x and y directions with steps Δx and Δy , and time is discretised into instants separated by a time step Δt . Consequently, each point in space is identified by integer indices (i, j) , and time by an index n . The continuous function $u(x, y, t)$ is replaced by a matrix of values $u_{i,j}^n$, representing the value of the function at the grid point (i, j) at time step n . The derivatives appearing in the differential equation are approximated using finite differences. In particular, both central and forward difference methods are implemented in the code and can be tested separately.

3 Assignment 2

4 Assignment 3